Subject Description Form

Subject Code	AP10011
Subject Title	Physics for Chemical and Biological Sciences
Credit Value	3
Level	1
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This is a subject designed for students in chemical and biological disciplines with no background in physics studies. Fundamental concepts in major physics topics (mechanics, heat, wave, optics and electricity) relevant to biology and chemistry will be discussed. The aim of this subject is to equip students with basic physics knowledge and appreciate the applications in Biology, Chemistry and Food Science.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) solve simple problems in kinematics, Newton's laws of motion, and energy; (b) solve problems in mechanics of many-particle systems; (c) solve simple problems in heat capacity and latent heat; (d) apply the superposition of waves; (e) explain phenomena related to the wave character of light; and (f) understand electrostatic field and potential;
Subject Synopsis/ Indicative Syllabus	**Mechanics**: scalars and vectors; kinematics and dynamics; Newton's laws; momentum, impulse, work and energy; conservation law; systems of particles; collisions; circular motion, rigid body rotation; angular momentum; conservation of momentum and conservation of energy. **Thermal physics**: heat and internal energy; heat capacity; conduction, convection and radiation; latent heat. **Waves and Optics**: nature of waves; wave motion; reflection and refraction; image formation by mirrors and lenses; superposition of waves; standing waves; diffraction and interference; electromagnetic spectrum; sound waves. **Electricity**: charges; Coulomb's law; electric field and potential.
Teaching/Learning Methodology	**Lecture**: Fundamentals in mechanics, thermal physics, waves, optics and electricity will be explained. Examples will be used to illustrate the concepts and ideas in the lecture. Students are free to request help. Homework problem sets will be given. **Student-centered Tutorial**: Students will work on a set of problems in tutorials. Students are encouraged to solve problems and to use their own knowledge to verify their solutions before seeking assistance. These problem sets provide them opportunities to apply their knowledge gained from the lecture. They also help the students to consolidate what they have learned. Furthermore, students can develop a deeper understanding of the subject in relation to daily life phenomena or experience. **e-learning**: In order to enhance the effectiveness of teaching and learning processes, electronic means and multimedia technologies would be adopted for presentations of

<table>
<tr><td></td><td>lectures (such as recorded lectures, open-source textbooks, demonstration videos), communication between students and lecturer (uReply, Kahoot!); delivery of handouts, homework and notices (remote-controlled experiments, learning management system) etc.</td></tr>
<tr><td>Assessment Methods in Alignment with Intended Learning Outcomes</td><td>
<table>
<tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="6">Intended subject learning outcomes to be assessed
(Please tick as appropriate)</td></tr>
<tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr>
<tr><td>(1) Continuous assessment</td><td>40</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>(2) Examination</td><td>60</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr>
<tr><td>Total</td><td>100</td><td colspan="6"></td></tr>
</table>
Continuous assessment:
The continuous assessment includes assignments and quizzes and test(s) which aim at checking the progress of students study throughout the course, assisting them in fulfilling the learning outcomes. Weighting distributions will be duly announced to students at the beginning of Semester.
Assignments in general include end-of-chapter problems, which are used to reinforce and assess the concepts and skills acquired by the students; and to let them know the level of understanding that they are expected to reach.
At least one test would be administered during the course of the subject as a means of timely checking of learning progress by referring to the intended outcomes, and as means of checking how effective the students digest and consolidate the materials taught in the class.

Examination: This is a major assessment component of the subject. It would be a closed-book examination. Complicated formulas would be given to avoid rote memory, such that the emphasis of assessment would be put on testing the understanding, analysis and problem solving ability of the students.</td></tr>
<tr><td>Student Study Effort Expected</td><td>
<table>
<tr><td>Class contact:</td><td></td></tr>
<tr><td>• Lecture</td><td>33 h</td></tr>
<tr><td>• Tutorial</td><td>6 h</td></tr>
<tr><td>Other student study effort:</td><td></td></tr>
<tr><td>• Self-study</td><td>81 h</td></tr>
<tr><td>Total student study effort</td><td>120 h</td></tr>
</table></td></tr>
<tr><td>Reading List and References</td><td>John D. Cutnell & Kenneth W. Johnson, Cutnell & Johnson Physics, 11th Edition, Asia Edition 2019, John Wiley & Sons.

Ruslan P. Ozerov & Anatoli A. Vorobyev, Physics for Chemists, 2007, Elsevier Science.

College Physics, Openstax (https://openstax.org/details/books/college-physics)

Hewitt, Conceptual Physics, 11th edition, 2010, Benjamin Cummings.

Radi, Hafez A., and John O. Rasmussen. Principles of Physics for Scientists and Engineers. Berlin ; New York: Springer, 2013. Undergraduate Lecture Notes in Physics. Web.</td></tr>
</table>